Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Sung-Min Cho	Email	tipa@jhu.edu
Institute Name	Johns Hopkins University			
Address	1800 Orleans Street, Baltimore, MD 21287			

2. Technology Overview

Title of Proposed Technology	AI-Driven Multimodal Pain Prediction Platform for Post-Neurosurgical Intensive Care Unit Patients			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices	Software	Software Solutions
Concept of Proposed Technology	The proposed technology is an **AI-driven multimodal pain prediction platform** designed to deliver **continuous, objective, and real-time assessment of severe pain** in adult post-neurosurgical intensive care unit (ICU) patients. The platform integrates **high-resolution physiological waveform data**—including electrocardiography, photoplethysmography, arterial blood pressure, and ventilator signals—with **electronic health record (EHR) contextual information** to generate a quantitative Pain Index. Using a deep learning–based model architecture, the system predicts severe pain events **30–60 minutes in advance**, enabling early clinical intervention. By replacing conventional intermittent bedside assessments with continuous monitoring, the platform supports **proactive analgesic titration** during the high-risk 24–48 hour postoperative window. The proposed project focuses on multi-institutional validation, Korea–U.S. technical alignment, and preparation for Software as a Medical Device (SaMD) regulatory pathways.			

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Rationale for Joint R&D Project	**Clinical and Technological Need** Pain assessment in adult ICU settings relies heavily on **subjective and episodic tools**, such as numeric rating scales and behavioral observation scores. These methods inadequately capture the dynamic nature of pain in critically ill patients, particularly those who are sedated, mechanically ventilated, delirious, or cognitively impaired. This limitation leads to delayed recognition of pain, reactive rather than proactive treatment, increased opioid exposure, higher risk of adverse events, and prolonged ICU stays. There is a clear unmet need for **objective, continuous, physiology-based pain monitoring technologies** that operate autonomously and reliably in complex ICU environments. **Rationale for International Joint Collaboration** Development and validation of an ICU-grade AI pain prediction platform require: • Access to **high-frequency, continuously acquired physiological waveform data** integrated with clinical context • **Multi-institutional datasets** to demonstrate generalizability and robustness • Expertise in **AI model development, clinical validation, and regulatory-aligned study design** International joint R&D enables federated learning–based collaboration across institutions without sharing raw patient data, supporting data privacy while enabling scalable model development. This structure integrates complementary expertise across Korea and the United States and ensures scientific credibility, regulatory readiness, and accelerated global deployment.
Key Words	1. Multimodal Biosignal-Based Pain Prediction 2. AI-Driven Clinical Decision Support System (CDSS) 3. Real-Time Continuous Pain Monitoring 4. Federated Learning for Medical AI 5. FDA/MFDS SaMD Validation Pipeline

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2)	**End of the Project**	Level (7)
R&D Objectives	**Overall Objective** To develop and validate an **AI-based, continuous pain prediction platform** for ICU patients and advance the technology to **TRL 7**, enabling readiness for real-world clinical validation and commercialization. **Specific Objectives** • **Construct standardized, de-identified multimodal datasets** integrating high-resolution ICU physiological waveforms with longitudinal electronic health record data across participating institutions. • **Identify and validate physiology-based biomarkers** associated with pain escalation and analgesic response in postoperative and ICU populations. • **Develop and externally validate a federated learning–based pain prediction model** that enables secure, cross-institutional model training without transfer of raw patient data. • **Demonstrate predictive performance superior to existing pain assessment and monitoring** approaches, including earlier detection of moderate-to-severe pain events.			

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	• **Establish a regulatory-aligned clinical and technical validation framework** to support Software as a Medical Device (SaMD) submission and downstream clinical deployment.
R&D Target Specifications	Key performance targets include: • **Prediction Discrimiation**: Achieve an area under the receiver operating characteristic curve (**AUROC ≥ 0.90**) for prediction of moderate-to-severe postoperative pain events. • **Sensitivity (Clinical Safety Threshold):** Demonstrate **≥ 95% sensitivity** for detection of severe pain events, prioritizing early identification to minimize false-negative risk in high-acuity postoperative and ICU populations. • **Prediction Horizon:** Provide reliable **30–60 minute advance prediction** of impending pain escalation, enabling anticipatory clinical intervention. • **Multimodal Data Integration and Harmonization**: Construct and validate **time-synchronized, standardized multimodal datasets** integrating high-resolution physiological waveforms (e.g., ECG, PPG, arterial blood pressure, ventilator signals) with longitudinal EHR data across institutions. • **External Validation and Clinical Utility**: Demonstrate **robust clinical validation**, including calibration analysis and decision-curve analysis, to confirm generalizability, clinical utility, and readiness for regulatory evaluation.
R&D Plan	The project will be conducted over **36 months** using a structured three-phase approach. **Phase 1: Retrospective Data Infrastructure and Model Development (Months 1–12)** • Establish regulatory approvals and standardized databases • Construct retrospective multimodal datasets • Develop physiological feature extraction pipelines • Design and train initial AI model architectures • **Phase 1 Deliverables**: Curated multimodal datasets; validated feature pipelines; baseline AI models. **Phase 2: Model Advancement and Prospective Study Preparation (Months 13–24)** • Validate feature–pain associations using retrospective data • Complete first-generation model training and internal validation • Secure approvals and prepare for prospective data collection • **Phase 2 Deliverables**: Internally validated models; finalized study protocols; regulatory-ready prospective study framework **Phase 3: Multi-Institutional Prospective Validation (Months 25–36)** • Conduct external validation using independent datasets • Perform calibration, utility, and cross-cohort analyses • Complete multicenter observational validation studies • Finalize a commercialization-ready prototype at TRL 7 • **Phase 3 Deliverables**: Externally validated AI platform; clinical utility evidence; TRL 7–ready prototype.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will provide **clinical and scientific leadership**, including: • **Provision of High-Fidelity Clinical Data**: Enable access to large-scale, high-resolution ICU physiological waveform data integrated with longitudinal clinical and electronic health record information, in compliance with institutional and regulatory requirements.

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	• **Leadership in Clinical Study Design and Validation**: Lead the design of retrospective and prospective clinical studies, including definition of endpoints, validation metrics, and multi-institutional evaluation strategies aligned with regulatory expectations. • **Execution of Clinical Studies:** Conduct and oversee retrospective analyses and prospective observational validation studies in postoperative and ICU populations, ensuring data quality, protocol adherence, and rigorous outcome assessment. • **Scientific Expertise and Methodological Oversight:** Contribute domain expertise in critical care medicine, biosignal processing, and clinical AI evaluation to guide model development, interpretation, and refinement. • **Regulatory and SaMD Readiness Support**: Provide guidance on regulatory-aligned research practices and documentation to support Software as a Medical Device (SaMD) readiness under both U.S. FDA and Korean MFDS frameworks.
Expected R&R of Korean SME Partner	The Korean SME partner will serve as the industry lead and primary driver of product development and commercialization, with responsibilities including: • **AI Algorithm and System Architecture Development**: Lead the design, implementation, and optimization of AI algorithms and overall system architecture to ensure scalable, robust, and clinically deployable performance. • **End-to-End Platform and System Integration**: Develop and integrate hardware and software components into a cohesive, end-to-end pain prediction platform suitable for real-world clinical environments. • **Regulatory Execution and SaMD Approval Strategy**: Lead regulatory execution activities to support Software as a Medical Device (SaMD) approval pathways in both Korea (MFDS) and the United States (FDA), including preparation of technical documentation and regulatory submissions. • **Quality Management System (QMS) Establishment**: Establish and maintain compliant quality management systems (e.g., ISO 13485–aligned processes) to support product development, verification, validation, and post-market readiness. • **Commercialization and Global Market Entry Planning**: Develop and execute commercialization strategies, including market positioning, reimbursement considerations, strategic partnerships, and plans for global market entry.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** • Establishment of a continuous, objective, AI-based pain prediction platform capable of real-time operation in postoperative and ICU environments. • Identification and validation of physiology-based pain biomarkers with demonstrated clinical relevance and generalizability. • Development of a Software as a Medical Device (SaMD)–ready prototype achieving TRL 7, suitable for real-world clinical validation and regulatory submission planning. **Clinical Impact** • Improved recognition of pain and proactive, data-driven analgesic management in postoperative and intensive care settings. • Reduction in opioid-related adverse events and avoidance of unnecessary diagnostic or rescue interventions through earlier and more precise pain detection. • Enhanced patient safety and quality of care for non-communicative, sedated, or high-risk patient populations, including ICU and surgical patients. **Industrial and Economic Impact**

	• Creation of a globally competitive digital health solution addressing a major unmet need in pain management, with extensibility beyond postoperative pain to broader acute and critical care settings. • Strengthening of Korea's leadership in AI-driven medical device innovation through validated, export-ready SaMD technology. • Establishment of a scalable Korea–U.S. joint R&D and commercialization framework, enabling efficient translation from academic innovation to global markets. **Societal Impact** • Contribution to improved quality, safety, and efficiency of critical care delivery, with potential downstream reductions in healthcare utilization and cost. • Acceleration of precision medicine adoption in intensive care environments, supporting more individualized, data-driven clinical decision-making. • By supporting improved opioid stewardship and addiction-aware pain management, this project has the potential to reduce opioid-related harm and enhance patient safety in both the United States and South Korea, generating sustained positive community- and society-level impact through safer prescribing practices, reduced preventable complications, and improved public health outcomes.

6. References

(1) Biosignal-Based Pain Datasets and Multimodal Research

Cascella, M., et al.,"Artificial Intelligence for Automatic Pain Assessment: Current Approaches and Challenges," Journal of Clinical Monitoring and Computing, Vol. 37, No. 2, pp. 345–359, February 2023.

Rationale and Relevance:
This peer-reviewed review article provides a comprehensive overview of recent advances in AI-based automated pain assessment, highlighting the clinical need for objective pain monitoring, particularly in ICU and non-communicative patient populations. The study systematically analyzes the limitations of existing subjective pain assessment methods and outlines key technical challenges in current approaches.

Relevance to PainX:
This publication serves as a core theoretical reference supporting the clinical necessity and technological gapaddressed by PainX, particularly in the context of biosignal-based, objective pain assessment.

(2) Survey of Machine Learning–Based Pain Detection Research

Gouverneur, P., Badura, A., Li, F., et al.,"An Experimental and Clinical Physiological Signal Dataset for Automated Pain Recognition,"Scientific Data, Vol. 11, Article 328, pp. 1–18, July 2024.

Rationale and Relevance:
This study presents a rigorously curated physiological signal datasetfor automated pain recognition, constructed from both experimental and real-world clinical environments. The dataset includes multimodal biosignals such as ECG and electrodermal activity (EDA)and is openly released to support reproducible research.

Relevance to PainX:
The work closely aligns with PainX's multimodal biosignal-based pain prediction strategy, serving as an international benchmark for data standardization, model validation, and clinical applicability.